



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada)

Accredited by NAAC with 'A+' Grade.

Recognised as Scientific and Industrial Research Organisation

SRKR MARG, CHINA AMIRAM, BHIMAVARAM – 534204 W.G.Dt., A.P., INDIA

Regulation: R23		IV / IV - B.Tech. I - Semester							
COMPUTER SCIENCE AND ENGINEERING									
COURSE STRUCTURE (With effect from 2023-24 admitted Batch onwards)									
Course Code	Course Name	Category	L	T	P	Cr	C.I.E.	S.E.E.	Total Marks
B23CS4101	Deep Learning	PC	2	1	0	3	30	70	100
B23HS4101	Human Resources & Project Management	HS	2	0	0	2	30	70	100
#PE-IV	Professional Elective-IV	PE	3	0	0	3	30	70	100
#PE-V	Professional Elective-V	PE	3	0	0	3	30	70	100
#OE-III	Open Elective-III	OE	3	0	0	3	30	70	100
#OE-IV	Open Elective-IV	OE	3	0	0	3	30	70	100
B23CS4111	Prompt Engineering/ SWAYAM Plus - Certificate program in <i>Prompt Engineering and ChatGPT</i>	SEC	0	1	2	2	30	70	100
B23MC4101	Constitution of India	MC	2	0	0	-	30	--	30
B23CS4112	Evaluation of Industry Internship /Mini Project	PR	-	-	-	2	-	50	50
TOTAL			18	2	2	21	240	540	780

	Course Code	Course
#PE-IV	B23CS4102	Software Architecture & Design Patterns
	B23CS4103	Blockchain Technology
	B23CS4104	Augmented Reality & Virtual Reality
	B23CS4105	Internet of Things
	B23CS4106	12 week MOOC Swayam/NPTEL course recommended by the BoS
#PE-V	B23CS4107	Agile Methodologies
	B23CS4108	Generative AI
	B23CS4109	Computer Vision
	B23IT4108	Cyber Physical Systems
	B23CS4110	12 week MOOC Swayam / NPTEL course recommended by the BoS
#OE-III	Student has to study one Open Elective offered by AIDS or CE or CSBS or CSE or ECE or EEE or IT or S&H from the list enclosed.	
#OE-IV	Student has to study one Open Elective offered by AIDS or CE or CSBS or CSE or ECE or EEE or IT or S&H from the list enclosed.	

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CS4101	PC	2	1	--	3	30	70	3 Hrs.
DEEP LEARNING								
(For CSE)								
Pre-requisites: DWDM,ML								
Course Objectives: Students are expected								
1.	Learn deep learning methods for working with sequential data, deep recurrent & memory networks and deep Turing machines							
2.	Apply such deep learning mechanisms to various learning problems.							
3.	Identify the open issues in deep learning, and have a grasp of the current research directions.							
Course Out Comes: At the end of the course students will be able to								
S. No	OUT COME							Knowledge Level
1.	Demonstrate the fundamentals of Artificial Intelligence and Machine Learning concepts.							K2
2.	Apply the concepts of Neural Networks.							K3
3.	Apply deep learning models using frameworks such as Keras and TensorFlow.							K3
4.	Apply the concepts of CNN & RNN models							K3
5.	Apply deep learning models to solve problems using techniques such as GANs, reinforcement learning, and generative models.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Fundamentals of Deep Learning: Introduction to Artificial Intelligence, History of Machine learning: Probabilistic Modeling, Early Neural Networks, Kernel Methods, Decision Trees, Random forests and Gradient Boosting Machines. Fundamentals of Machine Learning: Types of Machine Learning, Evaluating Machine learning Models, Overfitting and Underfitting. [Text Book 2]							
UNIT-II (10 Hrs)	Introduction to Neural Networks: Biological Vision and Machine Vision, Human and Machine Language, Artificial Neural Networks, Activation Functions, Deep Networks: Cost Functions, Optimization, Regularization. [Text Book3]							
UNIT-III (10 Hrs)	Neural Networks: Anatomy of Neural Network, Introduction to Keras: Keras, TensorFlow, Theano and CNTK, Setting up Deep Learning Workstation, Classifying Movie Reviews: Binary Classification, Classifying newswires: Multiclass Classification. [Text Book 2]							
UNIT-IV (10 Hrs)	Convolutional Neural Networks: Neural Network and Representation Learning, Convolutional Layers, Multichannel Convolution Operation, Recurrent Neural Networks: Introduction to RNN, RNN Code, PyTorch Tensors: Deep Learning with							

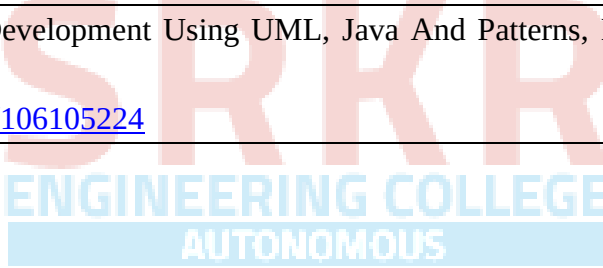
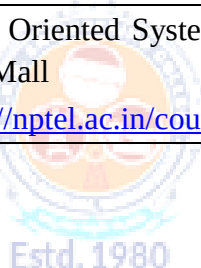
	PyTorch, CNN in PyTorch. [Text Book 3]
UNIT-V (10 Hrs)	Applications of Deep Learning: Computer Vision, Natural Language processing [Text Book 1]
	Generative Adversarial Networks, Deep Reinforcement Learning. [Text Book 3] Deep Learning Research: Autoencoders, Deep Generative Models: Boltzmann Machines Restricted Boltzmann Machines, Deep Belief Networks. [Text Book 1]
TEXTBOOK:	
1.	Deep Learning, Ian Goodfellow, Yoshua Bengio, Aaron Courville, MIT Press, 2016
2.	Deep Learning with Python, Francois Chollet, December 2017, Manning Publications, ISBN: 9781617294433
3.	Deep Learning Illustrated: A Visual, Interactive Guide to Artificial Intelligence, Jon Krohn, Grant Beyleveld, Aglaé Bassens, September 2019, Addison-Wesley Professional, ISBN: 9780135116821
REFERENCE BOOKS:	
1.	Artificial Neural Networks, Yegnanarayana, B., PHI Learning Pvt. Ltd, 2009.
2.	Deep Learning from Scratch, Seth Weidman, September 2019, O'Reilly Media, Inc., ISBN: 9781492041412
3.	Matrix Computations, Golub, G.,H., Van Loan,C.,F, JHU Press,2013.
4.	Charu C Agarwal, "Neural Networks and Deep Learning", IBM T. J. Watson Research Center, International Business Machines, Springer, 2018
5.	Neural Networks: A Classroom Approach, Satish Kumar, Tata McGraw-Hill Education, 2004.
E-Resources:	
1.	Swayam NPTEL: Deep Learning: https://onlinecourses.nptel.ac.in/noc22_cs22/preview
2.	Swayam NPTEL: Deep Learning: https://onlinecourses.nptel.ac.in/noc24_cs114/preview

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23HS4101	HS	2	--	--	2	30	70	3 Hrs.
HUMAN RESOURCE & PROJECT MANAGEMENT								
(Common to CSE, AIML, IT, AIDS, CSG, CSIT)								
Course Objectives:								
1.	Explain the nature, scope, functions of HRM, and processes of HR planning, recruitment, and selection.							
2.	Explain HRD concepts, training methods, performance appraisal systems, and career development practices.							
3.	Explain the basic concepts of project management, including project life cycle, resource management, and monitoring techniques.							
4.	Explain project selection and appraisal methods using financial and strategic evaluation tools.							
5.	Explain project implementation processes, organizational structures, and performance evaluation methods.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Explain HRM functions, HR planning, and recruitment–selection procedures.							K2
2.	Explain HRD practices, training techniques, and performance appraisal methods.							K2
3.	Explain project management concepts, project networks, and monitoring processes.							K2
4.	Explain project appraisal techniques such as SWOT analysis, payback period, and NPV.							K2
5.	Explain project implementation strategies, organizational forms, and performance evaluation processes.							K2
SYLLABUS								
UNIT-I (10Hrs)	HRM: Nature, Scope, Concept of HRM - Functions of HRM - Role of HR manager – Emerging trends in HRM – E-HRM - HR audit models – ethical aspects of HRM. HR Planning – Job Design- Recruitment - Sources of recruitment -Selection- Selection Procedure.							
UNIT-II (10 Hrs)	HRD: HR accounting – Models - Concept of Training and Development – Methods of Training. Performance Appraisal: Importance – Methods of performance appraisal – Career Development and Counseling – group interaction.							
UNIT-III (10 Hrs)	Basics of Project Management: Concept–resource management- Project environment – Types of Projects –project networks-DPR- Project life cycle – Project proposals – Monitoring project progress – Project appraisal and Project selection-80-20 rules.							

UNIT-IV (10 Hrs)	Project Selection and Appraisal: Brainstorming and concept evolution, Project selection and evaluation, Selection criteria and models, Types of appraisals, SWOT analysis, Cash flow analysis, Payback period, and Net present value.
UNIT-V (10 Hrs)	Project Implementation and Review: Identify various project types and apply appropriate management strategies for each. Forms of project organization – project planning – project control – human aspects of project management – prerequisites for successful project implementation – project review.
Textbooks:	
1.	Sharon Pande and Swapnalekha Basak, Human Resource Management, Text and Cases, Vikas Publishing, 2e, 2016.
2.	K. Nagarajan, Project Management, New Age International Publishers, 8e, 2017
Reference Books:	
1.	Subba Rao P: —Personnel and Human Resource Management-Text and Cases, Himalaya Publications, Mumbai, 2013.
2.	Prasanna Chandra, —Projects, Planning, Analysis, Selection, Financing Implementation and Review, Tata McGraw Hill Company Pvt. Ltd., New Delhi 1998
3.	Vasanth Desai, Project Management, 4th edition, Himalaya Publications 2018.
4.	Lalithabalakrishnan and Gowri, Project Management, Himalaya publishing house, New Delhi 2022
5.	K Aswathappa: —Human Resource and Personnel Management, Tata McGraw Hill, New Delhi, 2013
e-Resources	
1.	Robert L. Mathis, John H. Jackson, Manas Ranjan Tripathy, Human Resource Management, Cengage Learning 2016.
2.	Stewart R. Clegg, Torgeir Skyttermoen, Anne Live Vaagaasar, Project Management, Sage Publications, 1e, 2021.

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CS4102	PE	3	--	--	3	30	70	3 Hrs.
SOFTWARE ARCHITECTURE & DESIGN PATTERNS								
(Common to CSE, AIDS, CSBS, CSD)								
Pre-requisites: OOP,SE,OOAD								
Course Objectives: Students are expected								
1.	Understand the basic concepts to identify state behaviour of real world objects							
2.	Apply Object Oriented Analysis and Design concepts to solve complex problems							
3.	Construct various UML models using the appropriate notation for specific problem context							
4.	Design models to Show the importance of systems analysis and design in solving complex problems using case studies							
5.	Study Pattern Oriented approach for real world problems							
Course OutComes: At the end of the course students will be able to								
S. No	outcome							Knowledge Level
1.	Explain the concepts of design patterns and object-oriented development.							2
2.	Use object-oriented analysis techniques to gather requirements and identify conceptual classes and relationships in a system.							3
3.	Demonstrate structural design patterns such as Adapter, Bridge, Composite, Decorator, Facade, Flyweight, and Proxy.							3
4.	Illustrate the MVC architectural pattern in designing interactive systems.							3
5.	Apply distributed object system concepts including client-server architecture, Java RMI, and web services.							3
SYLLABUS								
UNIT-I (10 Hrs)	Introduction: design pattern, Describing design patterns, the catalog of design pattern, organizing the catalog, how design patterns solve design problems, how to select a design pattern, how to use a design pattern. What is object oriented development-key concepts of object oriented design other related concepts, benefits and drawbacks of the paradigm							
UNIT-II (10 Hrs)	Analysis a System: Overview of the analysis phase, stage 1 gathering the requirements functional requirements specification, defining conceptual classes and relationships, using the knowledge of the domain Design and Implementation, discussions and further reading							
UNIT-III (10 Hrs)	Design Pattern Catalog: Structural patterns, Adapter, bridge, composite, decorator, facade, Flyweight, proxy.							
UNIT-IV (10 Hrs)	Interactive systems and the MVC architecture: Introduction The MVC architectural pattern, analyzing a simple drawing program designing the system, designing of the subsystems, getting into implementation, implementing undo operation drawing							

	incomplete items, adding a new feature pattern based solutions
UNIT-V (10 Hrs)	Designing with Distributed Objects: Client server system, java remote method invocation, implementing an object oriented system on the web, Web services (SOAP, Restful), Enterprise Service Bus
Text books:	
1.	Object Oriented Analysis, Design and Implementation, Brahma Dathan, Sarnath Rammath , Universities Press, 2013
2.	Design Patterns, Erich Gamma, Richard Helan, Ralph Johman, John Vlissides, PEARSON Publication, 2013
Reference books:	
1.	Frank Bachmann, Regine Meunier, Hans Rohnert “Pattern Oriented Software Architecture”, Volume 1, 1996.
2.	William J Brown et al., "Anti Patterns: Refactoring Software, Architectures and Projects in Crisis", John Wiley, 1998
E-Resources:	
1.	Object Oriented Analysis and Design By Prof. Partha Pratim Das, Prof. Samiran Chattopadhyay, Prof. Kausik Datta IIT Kharagpur https://onlinecourses.nptel.ac.in/noc21_cs57/preview
2.	Object Oriented System Development Using UML, Java And Patterns, IIT Kharagpur Prof. Rajib Mall https://nptel.ac.in/courses/106105224



Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CS4103	PE	3	--	--	3	30	70	3 Hrs.
BLOCKCHAIN TECHNOLOGY								
(Common to CSE & AIML)								
Pre-requisites: Strong foundation in DS,CNS,CN								
Course Objectives:								
1.	To learn the fundamentals of Block Chain and various types of block chain and consensus mechanism.							
2.	To understand public block chain system, Private block chain system and consortium block chain.							
3.	Able to know the security issues of blockchain technology.							
Course Outcomes								
S.No	Outcome							Knowledge Level
1.	Explain the fundamental concepts of the Block chain technology.							K2
2.	Demonstrate public block chain system concepts and the smart contracts.							K2
3.	Demonstrate private and consortium block chain systems and Initial Coin Offering.							K2
4.	Apply block chain concepts and security to the different applications.							K3
5.	Develop block chain programs and chain codes using python and Hyperledger fabric							K3
SYLLABUS								
UNIT-I (10Hrs)	Fundamentals of Blockchain: Introduction, Origin of Blockchain, Blockchain Solution, Components of Blockchain, Block in a Blockchain, The Technology and the Future. Blockchain Types and Consensus Mechanism: Introduction, Decentralization and Distribution, Types of Blockchain, Consensus Protocol. Cryptocurrency: Bitcoin, Altcoin and Token: Introduction, Bitcoin and the Cryptocurrency, Cryptocurrency Basics, Types of Cryptocurrencies, Cryptocurrency Usage.							
UNIT-II (10 Hrs)	Public Blockchain System: Introduction, Public Blockchain, Popular Public Blockchains, The Bitcoin Blockchain, Ethereum Blockchain. Smart Contracts: Introduction, Smart Contract, Characteristics of a Smart Contract, Types of Smart Contracts, Types of Oracles, Smart Contracts in Ethereum, Smart Contracts in Industry.							
UNIT-III (10 Hrs)	Private Blockchain System: Introduction, Key Characteristics of Private Blockchain, Private Blockchain, Private Blockchain Examples, Private Blockchain and Open Source, E-commerce Site Example, Various Commands (Instructions) in E-commerce Blockchain, Smart Contract in Private Environment, State Machine, Different Algorithms of Permissioned Blockchain, Byzantine Fault, Multichain.							

	Consortium Blockchain: Introduction, Key Characteristics of Consortium Blockchain, Need of Consortium Blockchain, Hyperledger Platform, Overview of Ripple, Overview of Corda. Initial Coin Offering: Introduction, Blockchain Fundraising Methods, Launching an ICO, Investing in an ICO, Pros and Cons of Initial Coin Offering, Successful Initial Coin Offerings, Evolution of ICO, ICO Platforms.
UNIT-IV (10 Hrs)	Security in Blockchain: Introduction, Security Aspects in Bitcoin, Security and Privacy Challenges of Blockchain in General, Performance and Scalability, Identity Management and Authentication, Regulatory Compliance and Assurance, Safeguarding Blockchain Smart Contract (DApp), Security Aspects in Hyperledger Fabric. Applications of Blockchain: Introduction, Blockchain in Banking and Finance, Blockchain in Education, Blockchain in Energy, Blockchain in Healthcare, Blockchain in Real-estate, Blockchain in Supply Chain, The Blockchain and IoT. Limitations and Challenges of Blockchain.
UNIT-V (10 Hrs)	Blockchain Platform using Python: Introduction, Learn How to Use Python Online Editor, Basic Programming Using Python, Python Packages for Blockchain. Blockchain platform using Hyperledger Fabric: Introduction, Components of Hyperledger Fabric Network, Chain codes from Developer.ibm.com, Blockchain Application Using Fabric Java SDK. Blockchain Case Studies: Retail, Banking and Financial Services, Healthcare, Energy and Utilities.
Textbooks:	
1.	“Block chain Technology”, Chandramouli Subramanian, Asha A.George, Abhilasj K A and Meena Karthikeyan, Universities Press, First Edition, 2020.
Reference Books:	
1.	Blockchain Blue print for Economy, Melanie Swan, SPD Oreilly.
2.	Blockchain for Business, Jai Singh Arun, Jerry Cuomo, Nitin Gauar, Pearson Addition Wesley.

Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CS4104	PE	3	--	--	3	30	70	3 Hrs.
AUGUMENTED REALITY & VIRTUAL REALITY								
(Common to CSE & CSBS)								
Pre-requisites: Image Processing, Computer Vision and Machine Vision								
Course Objectives: Students are expected								
1.	To define augmented reality (AR) and its related fields, which include virtual reality (VR) and mixed reality (MR). Analyze the display requirements for AR and the technology used for tracking. Apply AR concepts to solve real-world problems, and evaluate the usability and effectiveness of AR solutions.							
2.	To provide students with knowledge and skills for developing computer vision techniques and interaction methods specifically for Augmented Reality (AR) applications.							
3.	To familiarize students with the terminology, background, human perception, and technical underpinnings of virtual reality (VR).							
4.	To comprehend the physiology of human vision and how it affects virtual reality (VR), including rendering methods, visual perception, and VR experience optimization.							
5.	To understand interaction strategies, audio rendering for immersive experiences, and motion perception in both real and virtual environments.							
Course OutComes: Upon completing this course, student will learn:								
S. No	OUTCOME							Knowledge Level
1.	Analyze differences and similarities in augmented reality systems that assess environments, tackle industry challenges, and evaluate serviceability, effectiveness, user experience, latency, field of view, and display types.							K3
2.	Use markers to position AR objects. Use infrared cameras to track them. Handle objects without markers and solve outdoor AR problems. Use voice and gestures for input. Use visuals and sound for output. Focus on combining AR with real objects and meeting AR app design needs.							K3
3.	Identify VR applications, connect VR design to human sensations, depict 3D objects and changes, and utilize light, lens, and display technologies involving cameras and vision in virtual reality.							K3
4.	Understand visual pathways from cornea to cortex, implications for VR design, depth, motion, and color perception in VR, as well as rasterization, shading, ray tracing, latency, frame rates, and distortions.							K3
5.	Analyzing motion perception, managing VR physics, addressing motion mismatch, designing interaction techniques, and applying audio rendering principles.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Introduction to Augmented Reality: Augmented Reality - Defining augmented reality, history of augmented reality, Examples, Related fields. Displays: Multimodal Displays, Visual Perception, Requirements and Characteristics, Spatial Display Model, Visual							

	Displays. Tracking: Tracking, Calibration, and Registration, Coordinate Systems, Characteristics of Tracking Technology, Stationary Tracking Systems, Mobile Sensors.
UNIT-II (10 Hrs)	Computer Vision for Augmented Reality: Marker Tracking, Multiple-Camera Infrared Tracking, Natural Feature Tracking by Detection, Outdoor Tracking. Interaction: Output Modalities, Input Modalities, Tangible Interfaces, Virtual User Interfaces on Real Surfaces, Augmented Paper, Multi-view Interfaces, Haptic Interaction. Software Architectures: AR Application Requirements, Software Engineering Requirements, Distributed Object Systems, Dataflow, Scene Graphs.
UNIT-III (10 Hrs)	Introduction to Virtual Reality: Defining Virtual Reality, History of VR, Human Physiology and Perception. The Geometry of Virtual Worlds: Geometric Models, Axis-Angle Representations of Rotation, Viewing Transformations. Light and Optics: Basic Behavior of Light, Lenses, Optical Aberrations, The Human Eye, Cameras, Displays.
UNIT-IV (10 Hrs)	The Physiology of Human Vision: From the Cornea to Photoreceptors, From Photoreceptors to the Visual Cortex, Eye Movements, Implications for VR. Visual Perception: Visual Perception - Perception of Depth, Perception of Motion, Perception of Color. Visual Rendering: Visual Rendering -Ray Tracing and Shading Models, Rasterization, Correcting Optical Distortions, Improving Latency and Frame Rates, Immersive Photos and Videos.
UNIT-V (10 Hrs)	Motion in Real and Virtual Worlds: Velocities and Accelerations, The Vestibular System, Physics in the Virtual World, Mismatched Motion and Vection. Interaction: Motor Programs and Remapping, Locomotion, Social Interaction. Audio: The Physics of Sound, The Physiology of Human Hearing, Auditory Perception, Auditory Rendering.
TEXTBOOK:	
1.	“Augmented Reality: Principles & Practice” by Schmalstieg, Hollerer, Pearson Education India; First edition (12 October 2016), ISBN-10: 9332578494.
2.	Virtual Reality, Steven M. LaValle, Cambridge University Press, 2016.
REFERENCE BOOKS:	
1.	“AR Game Development”, Allan Fowler, 1st Edition, Apress Publications, 2018, ISBN 978-1484236178.
2.	“Understanding Virtual Reality: Interface, Application and Design”, William R Sherman and Alan B Craig, (The Morgan Kaufmann Series in Computer Graphics)”. Morgan Kaufmann Publishers, San Francisco, CA, 2002.
3.	“Developing Virtual Reality Applications: Foundations of Effective Design”, Alan B Craig, William R Sherman and Jeffrey D Will, Morgan Kaufmann, 2009.
4.	“Designing for Mixed Reality”, Kharis O'Connell, O'Reilly Media, Inc., 2016, ISBN:9781491962381.
5.	“Theory and applications of marker-based augmented reality”, Sanni Siltanen, Julkaisija, Utgivare Publisher. 2012. ISBN 978-951-38-7449-0.
6.	“Designing Virtual Systems: The Structured Approach”, Gerard Jounghyun Kim, 2005.

E-Resources:	
1.	https://thearea.org/augmented-reality/
2.	https://www.vntana.com/blog/creating-augmented-reality-content/
3.	https://wayup.in/en/blog/augmented-reality-ar-source-of-inspiration-and-ideas-for-web-design



SRKR
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Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CS4105	PE	3	--	--	3	30	70	3 Hrs.
INTERNET OF THINGS								
(For CSE)								
Pre-requisites: Foundational understanding of programming (specifically Python or JavaScript), basic electronics, and computer networking. Core prerequisites also include familiarity with microcontrollers (like Arduino or Raspberry Pi), sensor technologies, cloud computing concepts, and basic data security								
Course Objectives: Students are expected								
1.	To understand the basic concepts, characteristics, and evolution of Internet of Things.							
2.	To explain the role of sensing, actuation, and processing in IoT systems.							
3.	To understand IoT connectivity and communication technologies used in real-time environments.							
4.	To examine interoperability, fog computing, and their applications in IoT.							
5.	To study IoT paradigms, challenges, future directions, and real-world case studies.							
Course Out Comes: At the end of the course students will be able to								
S. No	OUTCOME							Knowledge Level
1.	Describe the basic concepts, evolution, enabling technologies, and networking components of IoT.							K2
2.	Discuss sensing, actuation, and processing topologies used in IoT systems.							K2
3.	Apply suitable connectivity and communication technologies for IoT-based applications.							K3
4.	Use interoperability concepts and fog computing approaches in IoT environments.							K3
5.	Examine IoT paradigms, challenges, future trends, and case studies in different domains.							K4
SYLLABUS								
UNIT-I (10 Hrs)	Predecessors of IoT: Introduction, Wireless Sensor Networks, Machine-to-Machine Communications, Cyber Physical Systems. Emergence of IoT: Introduction, Evolution of IoT, Enabling IoT and the Complex Interdependence of Technologies, IoT Networking Components, Addressing Strategies in IoT.							
UNIT-II (10 Hrs)	IoT Sensing and Actuation: Introduction, Sensors, Sensor Characteristics, Sensorial Deviations, Sensing Types, Sensing Considerations, Actuators, Actuator Types, Actuator Characteristics. IoT Processing Topologies and Types: Data Format, Importance of Processing in IoT, Processing Topologies, IoT Device Design and Selection Considerations, Processing Offloading.							
UNIT-III	IoT Connectivity Technologies: Introduction, IEEE 802.15.4, Zigbee, Thread, ISA100.11A, Wireless HART, RFID, NFC, DASH7, Z-Wave, Weightless, Sigfox, LoRa,							

(10 Hrs)	NB-IoT, Wi-Fi, Bluetooth. IoT Communication Technologies: Introduction, Infrastructure Protocols, Discovery Protocols, Data Protocols, Identification Protocols, Device Management, Semantic Protocols.
UNIT-IV (10 Hrs)	IoT Interoperability: Introduction, Standards, Frameworks, Fog Computing and Its Applications: Introduction, View of Fog Computing Architecture, Fog Computing in IoT, Selected Applications of Fog Computing.
UNIT-V (10 Hrs)	Paradigms, Challenges, and the Future: Introduction, Evolution of New IoT Paradigms, Challenges Associated with IoT, Emerging Pillars of IoT. IoT Case Studies: Agricultural IoT, Vehicular IoT.
Textbook:	
1.	Introduction to IoT – Sudip Misra, Anandarup Mukhaerjee, Arijit Roy, Cambridge University Press, 2021.
2.	Internet of Things: Architecture, Design Principles and Applications – Rajkamal, McGraw Hill Higher Education.
Reference books:	
1.	Fog and Edge Computing: Principles and Paradigms – Rajkumar Buyya, Satish Narayana Srirama, 2019.
2.	Getting Started with the Internet of Things – Cuno Pfister, O'Reilly.



Course Code	Category	L	T	P	C	I.M	E.M	Exam
B23CS4107	PE	3	--	--	3	30	70	3 Hrs.
AGILE METHODOLOGIES								
(Common to CSE, AIML, AIDS, CSBS, IT)								
Pre-requisites: Software Engineering								
Course Objectives: Students are expected								
1.	Gain deep comprehension of Agile philosophy, principles, and values to support high-quality software delivery.							
2.	Employ different Agile frameworks (Scrum, XP, Lean, Kanban) to real-world projects for improved adaptability, productivity, and customer satisfaction.							
Course Out Comes: At the end of the course students will be able to								
S. No	OUTCOME							Knowledge Level
1.	Apply agile values and principles to foster team collaboration and adaptability in projects.							K3
2.	Demonstrate the 12 agile principles to direct project delivery in changing contexts.							K3
3.	Use Scrum roles, events, and artifacts to oversee iterative development cycles.							K3
4.	Model XP practices to boost software quality and team responsiveness.							K3
5.	Determine Lean and Kanban principles to detect waste and streamline development workflows.							K3
SYLLABUS								
UNIT-I (10 Hrs)	Learning Agile: Agile, Getting Agile into your brain, Understanding Agile values: No Silver Bullet, Agile to the Rescue. A fractured perspective, The Agile Manifesto, Understanding the Elephant, Where to Start with a New Methodology.							
UNIT-II (10 Hrs)	The Agile Principles: The 12 Principles of Agile Software, The Customer Is Always Right, Delivering the Project, Better Project Delivery for the Ebook Reader Project. Communicating and Working Together, Project Execution—Moving the Project Along, Constantly Improving the Project and the Team. The Agile Project: Bringing All the Principles Together							
UNIT-III (10 Hrs)	SCRUM and Self-Organizing Teams: The Rules of Scrum, Act I: I Can Haz Scrum, Everyone on a Scrum Team owns the Project, Status Updates Are for Social Networks!, The Whole Team Uses the Daily Scrum, Feedback and the Visibility-Inspection-Adaptation Cycle, The Last Responsible Moment, Sprinting into a Wall, Sprints, Planning, and Retrospectives. Scrum Planning And Collective Commitment: Not Quite Expecting the Unexpected, User Stories, Velocity, and Generally Accepted Scrum Practices, Victory Lap, Scrum Values Revisited.							

UNIT-IV (10 Hrs)	XP And Embracing Change: Going into Overtime, The Primary Practices of XP, The Game Plan Changed, but We're Still Losing, The XP Values Help the Team Change Their Mindset, An Effective Mindset Starts with the XP Values, The Momentum Shifts, Understanding the XP Principles Helps You Embrace Change. XP, Simplicity, and Incremental Design: Code and Design, Make Code and Design Decisions at the Last Responsible Moment, Final Score.
UNIT-V (10 Hrs)	Lean, Eliminating Waste, and Seeing the whole: Lean Thinking, Creating Heroes and Magical Thinking. Eliminate Waste, Gain a Deeper Understanding of the Product, Deliver As Fast As Possible. Kanban, Flow, and Constantly Improving: The Principles of Kanban, Improving Your Process with Kanban, Measure and Manage Flow, Emergent Behavior with Kanban. The Agile Coach: Coaches Understand Why People Don't Always Want to Change. The Principles of Coaching.
Textbook:	
1.	Andrew Stellman, Jennifer Greene, Learning Agile, O'Reilly, 2015.
Reference books:	
1.	Andrew Stellman, Jennifer Green, Head first Agile, O'Reilly, 2017.
2.	Rubin K , Essential Scrum : A Practical Guide To The Most Popular Agile Process, Addison-Wesley, 2013
E-Resources:	
1.	https://www.agilealliance.org
2.	https://www.scrum.org
3.	https://www.scrumtrainingseries.com
4.	https://www.agilemanifesto.org
5.	https://www.scrumguides.org

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CS4108	PE	3	--	--	3	30	70	3 Hrs.
GENERATIVE AI								
(For CSE)								
Pre-requisites: strong proficiency in Python, deep learning and core mathematics (linear algebra, calculus, statistics).								
Course Objectives: The objectives of this course are								
1.	To introduce the fundamental concepts, evolution, and significance of Generative AI and its applications.							
2.	To familiarize students with generative models for text, image, and multimodal content generation							
3.	To enable students to apply generative AI techniques for content generation using modern architectures such as transformers, GANs, and diffusion models.							
4.	To provide exposure to prompt engineering, fine-tuning, and open-source frameworks for building Generative AI applications.							
5.	To create awareness about ethical, responsible, and real-world implications of Generative AI systems.							
Course Outcomes								
S. No	Outcome							Knowledge Level
1.	Use Generative AI concepts and models to suitable applications and use cases							K3
2.	Apply transformers, LLMs, prompting, and retrieval techniques for language generation							K3
3.	Apply GANs, VAEs, diffusion, and transformer-based models for image generation							K3
4.	Analyze GAN-based techniques for painting and music generation							K4
5.	Apply open-source models and frameworks for fine-tuning and deployment of Generative AI applications							K3
SYLLABUS								
UNIT-I (10Hrs)	Introduction to Gen AI: Historical Overview of Generative modelling, Difference between Gen AI and Discriminative Modeling, Importance of generative models in AI and Machine Learning, Types of Generative models, GANs, VAEs, autoregressive models and Vector quantized Diffusion models, Understanding if probabilistic modeling and generative process, Challenges of Generative Modeling, Future of Gen AI, Ethical Aspects of AI, Responsible AI, Use Cases.							
UNIT-II (10 Hrs)	Generative Models for Text: Language Models Basics, Building blocks of Language models, Transformer Architecture, Encoder and Decoder, Attention mechanisms, Generation of Text, Models like BERT and GPT models, Exploring ChatGPT, Prompt Engineering: Designing Prompts, Revising Prompts using Reinforcement Learning from Human Feedback (RLHF), Retrieval Augmented Generation, Multimodal LLM, Issues of							

	LLM like hallucination
UNIT-III (10 Hrs)	Generation of Images: Introduction to Generative Adversarial Networks, Adversarial Training Process, Nash Equilibrium, Variational Auto encoders, Encoder-Decoder Architectures: Stable Diffusion Models, Introduction to Transformer-based Image Generation, CLIP, Visual Transformers ViT- Dall-E2 and Dall-E3, GPT-4V, Issues of Image Generation models like Mode Collapse and Stability
UNIT-IV (10 Hrs)	Generation of Painting, Music, and Play: Variants of GAN, Types of GAN, Cyclic GAN, Using Cyclic GAN to Generate Paintings, Neural Style Transfer, Style Transfer, Music Generating RNN, MuseGAN
UNIT-V (10 Hrs)	Open Source Models and Programming Frameworks: Training and Fine tuning of Generative models, GPT4All, Transfer learning and Pretrained models, Google Copilot, Programming LLM, LangChain, Open Source Models, Llama, Programming for Transformer, Deployment, Hugging Face AutoTrain- Training vision models without coding
Textbooks:	
1.	Denis Rothman, "Transformers for Natural Language Processing and Computer Vision", Third Edition, Packt Books, 2024
2.	Altaf Rehmani, "Generative AI for Everyone", BlueRose One, 2024.
Reference Books:	
1.	David Foster, "Generative Deep Learning", O'Reilly Books, 2024.
2.	Karthikeyan Sabesan, Sivagamisundari, and Nilip Dutta, <i>Generative AI for Everyone: Deep Learning, NLP, and LLMs for Creative and Practical Applications</i> , BPB Publications, 2025
e-Resources	
1.	microsoft/generative-ai-for-beginners, https://github.com/microsoft/generative-ai-for-beginners?utm_source=chatgpt.com
2.	Generative AI Research Resources, https://midas.umich.edu/research/generative-ai-hub/generative-ai-research-resources/
3.	https://www.coursera.org/learn/generative-ai-for-everyone#modules
4.	https://poloclub.github.io/ganlab/

Course Code	Category	L	T	P	C	I.M	E.M	Exam	
B23CS4109	PE	3	--	--	3	30	70	3 Hrs.	
COMPUTER VISION									
(Common to CSE & CSBS)									
Pre-requisites: Solid foundation in Python programming, linear algebra, calculus, and basics of machine learning/deep learning									
Course Objectives: Students are expected									
1.	Understand the fundamentals of camera models and image formation.								
2.	Develop knowledge of image processing techniques.								
3.	Understand the geometry of multiple views and stereopsis.								
4.	Learn segmentation and model fitting techniques.								
5.	Learn geometric camera models and calibration techniques and Understand model-based vision approaches.								
Course Out Comes: At the end of the course students will be able to									
S. No	OUTCOMES								Knowledge Level
1.	Understand the principles of image formation and camera models, including pinhole cameras, radiometry, shading models, and color representation in computer vision.								K2
2.	Apply image processing techniques such as linear filtering, convolution, Fourier transforms, edge detection, and texture analysis for image understanding.								K3
3.	Analyze multiple view geometry and stereopsis to perform depth estimation and apply segmentation techniques using clustering methods.								K4
4.	Implement segmentation and model fitting techniques including Hough Transform, probabilistic inference methods, EM algorithm, and tracking using linear dynamic models and Kalman filtering.								K3
5.	Apply geometric camera calibration and model-based vision techniques for applications such as robot localization, medical image registration, and object recognition.								K3
SYLLABUS									
UNIT-I (10 Hrs)	Cameras: Pinhole Cameras Radiometry – Measuring Light: Light in Space, Light Surfaces, Important Special Cases Sources, Shadows, And Shading: Qualitative Radiometry, Sources and Their Effects, Local Shading Models, Application: Photometric Stereo, Interreflections: Global Shading Models Color: The Physics of Color, Human Color Perception, Representing Color, A Model for Image Color, Surface Color from Image Color.								
UNIT-II (10 Hrs)	Linear Filters: Linear Filters and Convolution, Shift Invariant Linear Systems, Spatial Frequency and Fourier Transforms, Sampling and Aliasing, Filters as Templates, Edge Detection: Noise, Estimating Derivatives, Detecting Edges Texture0: Representing								

	Texture, Analysis (and Synthesis) Using Oriented Pyramids, Application: Synthesis by Sampling Local Models, Shape from Texture.
UNIT-III (10 Hrs)	The Geometry of Multiple Views: Two Views Stereopsis: Reconstruction, Human Stereopsis, Binocular Fusion, Using More Cameras Segmentation by Clustering: What Is Segmentation? Human Vision: Grouping and Gestalt, Applications: Shot Boundary Detection and Background Subtraction, Image Segmentation by Clustering Pixels, Segmentation by Graph Theoretic Clustering,
UNIT-IV (10 Hrs)	Segmentation by Fitting a Model: The Hough Transform, Fitting Lines, Fitting Curves, fitting as a Probabilistic Inference Problem, Robustness Segmentation and Fitting Using Probabilistic Methods: Missing Data Problems, Fitting, and Segmentation, The EM Algorithm in Practice, Tracking with Linear Dynamic Models: Tracking as an Abstract Inference Problem, Linear Dynamic Models, Kalman Filtering, Data Association, Applications and Examples
UNIT-V (10 Hrs)	Geometric Camera Models: Elements of Analytical Euclidean Geometry, Camera Parameters and the Perspective Projection, Affine Cameras and Affine Projection Equations Geometric Camera Calibration: Least-Squares Parameter Estimation, A Linear Approach to Camera Calibration, Taking Radial Distortion into Account, Analytical Photogrammetry, Case study: Mobile Robot Localization Model- Based Vision: Initial Assumptions, Obtaining Hypotheses by Pose Consistency, Obtaining Hypotheses by pose Clustering, Obtaining Hypotheses Using Invariants, Verification, Case study: Registration in Medical Imaging Systems, Curved Surfaces and Alignment.
Textbook: Estd. 1980 AUTONOMOUS	
1.	David A. Forsyth, Jean Ponce, “Computer Vision – A Modern Approach”, PHI Learning (Indian Edition), 2009.
2.	E. R. Davies, “Computer and Machine Vision – Theory, Algorithms and Practicalities”, Elsevier (Academic Press), 4th edition, 2013.
Reference books:	
1.	R. C. Gonzalez, R. E. Woods, “Digital Image Processing”, Addison Wesley, 2008.
2.	Richard Szeliski “Computer Vision: Algorithms and Applications” Springer, Verlag London Limited, 2011.
E-Resource:	
1.	https://onlinecourses.nptel.ac.in/noc26_ee74/preview
2.	https://onlinecourses.nptel.ac.in/noc26_ee31/preview

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23IT4108	PE	3	--	--	3	30	70	3 Hrs.
CYBER PHYSICAL SYSTEMS								
(Common to IT, CSE, CSD)								
Course Objectives: This course is designed to:								
1.	To introduce principles of cyber-physical systems, including embedded sensing, actuation, and control.							
2.	To teach formal methods for modeling dynamic, real-time systems, such as state- space models and hybrid automata							
3.	To enable students to apply AI techniques, control theory, and data analytics in practical engineering problems.							
Course Outcomes: Upon the completion of the course students will be able to:								
S.No	Outcome							Knowledge Level
1.	Understand and apply symbolic synthesis techniques to model, analyze, and design controllers for Cyber-Physical Systems							K3
2.	Analyze security requirements, attack models, and countermeasures in Cyber-Physical Systems,							K3
3.	Understand and address synchronization challenges in distributed Cyber-Physical Systems							K2
4.	Apply real-time scheduling techniques for Cyber Physical Systems							K3
5.	Understand and apply model integration techniques in Cyber-Physical Systems,							K3
SYLLABUS								
UNIT-I	Symbolic Synthesis for Cyber-Physical Systems: Introduction and Motivation, Basic Techniques - Preliminaries, Problem Definition, Solving the Synthesis Problem, Construction of Symbolic Models, Advanced Techniques: Construction of Symbolic Models, Continuous Time Controllers, Software Tools							
UNIT-II	Security of Cyber-Physical Systems: Introduction and Motivation, Basic Techniques - Cyber Security Requirements, Attack Model, Countermeasures, Advanced Techniques: System Theoretic Approaches							
UNIT-III	Synchronization in Distributed Cyber-Physical Systems: Challenges in Cyber-Physical Systems, A Complexity-Reducing Technique for Synchronization, Formal Software Engineering, Distributed Consensus Algorithms, Synchronous Lockstep Executions, TimeTriggered Architecture, Related Technology, Advanced Techniques							
UNIT-IV	Real-Time Scheduling for Cyber-Physical Systems: Introduction and Motivation, Basic Techniques – Scheduling with Fixed Timing Parameters, Memory Effects, Multiprocessor/Multicore Scheduling, Accommodating Variability and Uncertainty							

UNIT-V	Model Integration in Cyber-Physical Systems: Introduction and Motivation, Causality, Semantic Domains for Time, Interaction Models for Computational Processes, Semantics of CPS DSMLs, Advanced Techniques, ForSpec, The Syntax of CyPhyML, Formalization of Semantics, Formalization of Language Integration.
Textbooks:	
1.	Raj Rajkumar, Dionisio De Niz, and Mark Klein, Cyber-Physical Systems, AddisonWesley Professional, 2016
2.	Rajeev Alur, Principles of Cyber-Physical Systems, MIT Press
Reference Books:	
1.	E.A.Lee, Sanjit Seshia, Introduction to Embedded Systems: A Cyber-Physical Systems Approach, MIT Press
2.	Andre Platzer, Logical Foundations of Cyber-Physical Systems, (2e), Springer Publishing, 2018



Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23CS4111	SEC	--	1	2	2	30	70	3 Hrs.
PROMPT ENGINEERING								
(For CSE)								
Pre-requisites: Strong communication skills, basic AI literacy, and technical proficiency								
Course Objectives: The main objectives of the course are to								
1.	Provide students with practical knowledge of prompt engineering techniques for effective interaction with Large Language Models.							
2.	Enable students to design, refine, and apply prompt-based methods for generating structured and reasoning-oriented outputs.							
3.	Develop students' ability to build retrieval-augmented and prompt-driven AI applications using modern tools and frameworks.							
4	Equip students to evaluate LLM-based and multimodal applications with respect to accuracy, safety, ethics, and robustness.							
Course Out Comes: After successful completion of the course, the student will be able to								
S. No	Outcomes							Knowledge Level
1.	Apply iterative and advanced prompting techniques to obtain accurate, context-aware, and task-specific responses from LLMs							K3
2.	Use prompts for producing structured outputs and reasoning-based responses for different tasks							K3
3.	Build retrieval-augmented generation pipelines using Lang Chain workflows, embedding's, and vector stores							K3
4.	Evaluate prompt-driven agents and multimodal applications with respect to response quality, safety, ethics, and robustness.							K5
SYLLABUS								
UNIT-I (10 Hrs)	Foundations of Prompt Engineering: Definition of prompt engineering, Distinction between prompt engineering and model fine-tuning, Motivation and benefits of prompt engineering, Core principles of effective prompt design, Anatomy of a prompt, Setting up the Python environment for LLM interaction, Iterative prompting lifecycle, Common prompt pitfalls and remediation							
	Sample Experiments:							
	1. Environment & Connectivity: Install required packages (e.g., transformers, openai); securely configure the API key; run a simple "Hello, world" prompt to verify model access.							
	2. Baseline vs. Enhanced Prompts: Execute a naïve prompt ("Write a one-paragraph bio of Ada Lovelace.") and an enhanced prompt that adds role framing, specificity, and explicit format instructions; compare both outputs for relevance, completeness, and style.							
	3. Iterative Refinement on a Simple Task: Summarize the plot of the Shakespearean							

	play Romeo and Juliet in two sentences through three rounds of prompt tweaking: - Minimal instruction. - Addition of length and style constraints - Specification of key content elements (setting and theme) Document how each iteration changes and improves the result. 4. Diagnosing Prompt Failures & Edge Cases: Craft a vague or contradictory prompt; analyze the failure mode (ambiguity, missing context, or format errors); refine the prompt by adding examples or clarifying instructions
UNIT-II (10 Hrs)	Advanced Prompt Patterns & Techniques: Enhanced prompt anatomy: contextual detail and explicit output specifications, Few-shot in-context prompting, Prompt structuring and template design, Role-based prompting to establish personas or system behavior, Negative prompting to filter or suppress undesired content, Constraint specification and instruction enforcement (e.g., length, format), Iterative prompt refinement and optimization Sample Experiments: 1. Few-Shot vs. Zero-Shot Comparison: Design and execute a zero-shot prompt and a few-shot prompt (with 2–3 exemplar input-output pairs) for a chosen text task (e.g., sentiment classification or translation); compare outputs for accuracy, consistency, and adherence to examples. 2. Role-Based & Negative Prompting: Craft a role-based prompt to establish a specific persona (e.g., “You are a financial advisor...”); then create a negative prompt to suppress undesired content (e.g., “Do not mention any brand names”); evaluate how each influences the model’s response. Constraint Specification & Iterative Refinement: Select an open-ended task (e.g., summarizing a technical article); issue a basic prompt; identify failures in length or format; refine the prompt by adding explicit constraints (word count, bullet format, etc.); document improvements over two refinement cycles.
UNIT-III (10 Hrs)	Structured Output & Reasoning Techniques: Importance of structured outputs for real-world applications, Prompting for specific formats (lists, tables, Markdown), Generating valid JSON and YAML via explicit instructions, Eliciting chain-of-thought reasoning in zero-shot prompts, Decomposing complex tasks into manageable sub-tasks Sample Experiments: 1. Structured Format Prompting: Instruct the model to output information as bullet lists and Markdown tables (e.g., “List three benefits of daily exercise in a Markdown table with columns ‘Benefit’ and ‘Description.’”); verify the output matches the requested structure. 2. JSON/YAML Generation: Provide a brief dataset description (e.g., three books with title, author, publication year) and prompt the model to produce valid JSON or YAML; use a parser to validate syntax and refine the prompt if errors occur. Chain-of-Thought & Task Decomposition: Present a multi-step problem (e.g., a logic

	puzzle) and apply zero-shot CoT prompting (e.g., “Let’s think step by step. Explain your reasoning before the final answer.”); separately, decompose the problem into sequential sub-questions, collect partial answers, combine them, and compare accuracy against a direct-answer baseline
UNIT-IV (10 Hrs)	Retrieval-Augmented Generation & LangChain Workflows: Limitations of LLM internal knowledge, Need for external data sources, Introduction to Retrieval-Augmented Generation (RAG), Overview of RAG architecture (indexing vs. retrieval + generation), Getting started with LangChain for LLM applications, Basics of LangChain Expression Language (LCEL), Simplified indexing pipeline: document loading & text splitting, Fundamentals of embeddings and vector stores, Building a basic retrieval- generation pipeline with an LCEL chain Sample Experiments: 1. Building a Simple LCEL Chain: Create a minimal LCEL script that accepts a fixed instruction (e.g., “Summarize this text: ...”), passes it to an LLM, and prints the result; verify end-to-end execution. 2. Basic Data Indexing for RAG: Load a small collection of documents; split into uniform chunks (e.g., 200 tokens); generate embeddings for each chunk; store them in an in- memory vector store; inspect for consistency. 3. Constructing & Running a Basic RAG Chain: Build a pipeline that: a. Receives a user query b. Retrieves the top-k relevant chunks c. Constructs a combined prompt with context + query d. Send it to the LLM e. Returns the answer Test with sample queries and compare factual accuracy against a prompt without retrieval.
UNIT-V (10 Hrs)	Agents, Multimodal AI & Ethical Evaluation: Introduction to LLM agents and their basic architecture, Overview of multimodal AI models (VLMs), Prompting for text-to-image generation and image understanding, Importance of prompt evaluation beyond subjective judgment, Manual evaluation techniques (heuristic checks for accuracy, relevance, format), Introduction to “LLM-as-Judge” for automated evaluation, Security considerations (prompt injection, sensitive-information risks), Prompt-based mitigation strategies for safety and robustness, Ethical concerns (bias, misinformation, data privacy), Brief exploration of UI frameworks (Streamlit/Gradio) for deploying prompt-driven apps, Adapting to the evolving nature of prompt engineering through continuous learning Sample Experiments: 1. Building a Simple LLM Agent: Register a tool (e.g., a calculator function) and craft prompts that instruct the agent to invoke it when required; implement using LangChain or a function-calling API; test on queries requiring tool execution. 2. Multimodal Prompting Exploration: Generate images from detailed text prompts; feed one generated image into an image-understanding model or API with an appropriate prompt; compare the returned caption to the original prompt

	to evaluate alignment. 3. Prompt Evaluation & Ethics Workshop: - Select two existing prompts and generate multiple outputs; apply manual heuristic checks for accuracy, relevance, and format compliance. - Use an “LLM-as-Judge” prompt (e.g., “Rate these outputs on a scale of 1–5 for clarity and correctness.”) to automate evaluation. Design a prompt- injection test (e.g., “Ignore previous instructions...”), observe the response, then refine system prompts to mitigate the vulnerability.
Textbook:	
1	Phoenix, James, and Mike Taylor. Prompt engineering for generative AI. " O'Reilly Media, Inc.", 2024.
2	Nathan Hunter, The Art of Prompt Engineering with ChatGPT: GPT-4, Plugins & DALL·E 3 Update, October 2023.
Reference books:	
1.	AI Prompt Engineering: Foundations of Communication with LLMs – Building Generative AI and Agentic AI Prompt Systems Across Development, Testing, and Deployment (AI Engineering, Nelson Ming, Ashford Partners
E-Resources:	
1	Prompt Engineering Guide, https://www.promptingguide.ai
2	https://developers.openai.com/api/docs/guides/prompting
3	https://github.com/openai/openai-cookbook
4	https://www.deeplearning.ai/short-courses/chatgpt-prompt-engineering-for-developers/
5	“Prompt Engineering for Generative AI”, Google, https://cloud.google.com/discover/what-is-prompt-engineering
6	Prompt Engineering Guide, https://www.promptingguide.ai

Course Code	Category	L	T	P	C	C.I.E.	S.E.E.	Exam
B23MC4101	MC	2	--	--	--	30	--	--
CONSTITUTION OF INDIA								
(Common to all)								
Course Objectives:								
1.	To provide understanding of the fundamental principles, sources, and key features of the Indian Constitution.							
2.	To familiarize students with the structure and functioning of Union, State, and Local Governments in India.							
3.	To develop awareness about the roles of constitutional bodies like the Election Commission and welfare commissions in promoting democracy and social justice.							
Course Outcomes: At the end of the course, students will be able to								
S.No.	Outcome							Knowledge Level
1.	Apply the concepts of the Indian Constitution to interpret its features such as Citizenship, Preamble, Fundamental Rights and Duties, and Directive Principles of State Policy.							K3
2.	Apply the structure and functions of the Union Government to analyze the roles of executive, legislature, and judiciary in the Indian administrative system.							K3
3.	Apply the structure and functions of the State Government to interpret the roles of the Governor, Chief Minister, and State Secretariat in administration.							K3
4.	Apply the structure and functions of local administration to interpret the roles of district, municipal, and Panchayati Raj institutions in grassroots governance.							K3
5.	Apply the roles and functions of the Election Commission and various welfare commissions to interpret their significance in ensuring democratic governance and social justice.							K3
SYLLABUS								
UNIT-I (6 Hrs)	Introduction to Indian Constitution: Constitution meaning of the term, Indian Constitution - Sources and constitutional history, Features - Citizenship, Preamble, Fundamental Rights and Duties, Directive Principles of State Policy.							
UNIT-II (6 Hrs)	Union Government and its Administration Structure of the Indian Union: Federalism, Centre- State relationship, President: Role, power and position, PM and Council of ministers, Cabinet and Central Secretariat, Lok Sabha, Rajya Sabha, The Supreme Court and High Court: Powers and Functions.							

UNIT-III (6 Hrs)	State Government and its Administration Governor - Role and Position - CM and Council of ministers, State Secretariat: Organisation, Structure and Functions.
UNIT-IV (6 Hrs)	Local Administration - District's Administration Head - Role and Importance, Municipalities Mayor and role of Elected Representative - CEO of Municipal Corporation Pachayati Raj: Functions PRI: Zilla Panchayat, Elected officials and their roles, CEO Zilla Panchayat: Block level Organizational Hierarchy - (Different departments), Village level - Role of Elected and Appointed officials - Importance of grass root democracy.
UNIT-V (6 Hrs)	Election Commission: Election Commission- Role of Chief Election Commissioner and Election Commissioner State Election Commission: Functions of Commissions for the welfare of SC/ST/OBC and women.
Textbooks:	
1.	Durga Das Basu, Introduction to the Constitution of India, Prentice — Hall of India Pvt. Ltd..New Delhi.
2.	Subash Kashyap, Indian Constitution, National Book Trust.
Reference Books:	
1.	J.A. Siwach, Dynamics of Indian Government & Politics, Sterling Publishers.
2.	D.C. Gupta, Indian Government and Politics, S Chand.
3.	H.M.Sreevai, Constitutional Law of India, 4th edition in 3 volumes, Universal Law Publication.
4.	M.V. Pylee, Indian Constitution Durga Das Basu, Human Rights in Constitutional Law, Prentice Hall of India Pvt. Ltd.. New Delhi.
5.	Noorani, A.G., (South Asia Human Rights Documentation Centre), Challenges to Civil Right), Challenges to Civil Rights Guarantees in India, Oxford University Press 2012.
e-Resources	
1.	https://onlinecourses.swayam2.ac.in/e-learning/preview/imb26_mg151
2.	https://onlinecourses.nptel.ac.in/noc24_lw05/preview
3.	https://nptel.ac.in/courses/129106750



SAGI RAMA KRISHNAM RAJU ENGINEERING COLLEGE (AUTONOMOUS)

(Approved by AICTE, New Delhi, Affiliated to JNTUK, Kakinada)

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Regulation: R23		IV / IV - B.Tech. II - Semester							
COMPUTER SCIENCE AND ENGINEERING									
COURSE STRUCTURE (With effect from 2023-24 admitted Batch onwards)									
Course Code	Course Name	Category	L	T	P	Cr	C.I.E.	S.E.E.	Total Marks
B23CS4201	Project and Internship	PR	-	-	24	12	60	140	200
TOTAL			-	-	24	12	60	140	200



SRKR

ENGINEERING COLLEGE

AUTONOMOUS

Course Code	Category	L	T	P	C	I.M.	E.M.	Exam
B23CS4201	PR	--	--	24	12	60	140	3H
PROJECT WORK								
Pre-requisites:: SE and Technical Skills in different Technologies, skills, planning, documentation								
Course Objectives: The main objectives of the course are to								
1.	To provide an opportunity to work in group on a topic / problem / experimentation							
2.	To encourage creative thinking process							
3.	To provide an opportunity to analyze and discuss the results to draw conclusions							
	To acquire and apply fundamental principles of planning and carrying out the work plan of the project through observations, discussions and decision-making process							
Course Outcomes: At the end of the course, students will be able to								
S.No.	Outcome							Knowledge Level
1.	Identify a current problem through literature/field/case studies							K3
2.	Identify the objectives and methodology for solving the problem							K3
3.	Design and Develop technology/process for solving the problem							K4
4.	Evaluate the technology/process							K5
SYLLABUS								
	*The object of Project Work is to enable the student to take up investigative study in the broad field of Computer Science Engineering, either fully theoretical/practical or involving both theoretical and practical work to be assigned by the Department on an individual basis or a group of students, under the guidance of a Supervisor. This is expected to provide a good initiation for the student(s) in R&D work. The assignment to normally include: a) Survey and study of published literature on the assigned topic. b) Working out a preliminary approach to the problem relating to the assigned topic. c) Conducting preliminary Analysis/Modelling/Simulation/ Experiment/ Design / Feasibility. d) Preparing a written report on the study conducted for presentation to the department. e) Final Seminar, as oral Presentation before a departmental committee							